APPENDIX A
RDDP ALGORITHM UNDER RCR CONDITION

The systematic procedure of the conventional RDDP algorithm is summarized in Algorithm 1.

Algorithm 1 RDDP algorithm under RCR condition

Input: Given $\boldsymbol{\xi}_0$ and tolerance ϵ; initialize the convex hull $\{\boldsymbol{x}_t^j, \overline{\mathcal{Q}}_{t+1}(\boldsymbol{x}_t^j)\}$ and set $\underline{\mathcal{Q}}_{t+1}(\boldsymbol{x}_t) = -\infty$ for $t \in \{1, \ldots, T-1\}$; set the iteration index $z = 0$ and *flag* $\leftarrow 1$.

Output: $\underline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1})$ and $\overline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1}), \forall t \in \mathcal{T}$.

while *flag* **do**
  Solve the first-stage Problem (19) to obtain the state variables $\boldsymbol{x}_1$.
  if $\left|\left(\overline{\mathcal{Q}}_2(\boldsymbol{x}_1) - \underline{\mathcal{Q}}_2(\boldsymbol{x}_1)\right)/\overline{\mathcal{Q}}_2(\boldsymbol{x}_1)\right| \leqslant \epsilon$ **then**
    Set *flag* $\leftarrow 0$. ***// Terminate after current iteration***
  else
    $z = z + 1$
    Forward Pass:
    for $t \in \{2, \ldots, T\}$ **do**
      Let $\boldsymbol{\xi}_t^{\text{fw}}$ be the optimal solution of Problem (17).
      Derive the state decision $\boldsymbol{x}_t^z$ by solving Problem (19) with optimal value $\underline{Q}_t(\boldsymbol{x}_{t-1}^z, \boldsymbol{\xi}_t^{\text{fw}})$.
    end for
    Backward Pass:
    for $t \in \{T, \ldots, 2\}$ **do**
      Let $\boldsymbol{\xi}_t^z$ be the optimal solution of Problem (17). If $\overline{Q}_t(\boldsymbol{x}_{t-1}^z) < \overline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1}^z)$, add the point $\{\boldsymbol{x}_{t-1}^z, \overline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1}^z)\}$ to the convex hull constraint (17c) at stage $t-1$.
      Solve Problem (19) to obtain optimal value $\underline{Q}_t(\boldsymbol{x}_{t-1}^z, \boldsymbol{\xi}_t^z)$ and derive dual solution $\boldsymbol{\psi}_{t,z}$. If $\underline{Q}_t(\boldsymbol{x}_{t-1}^z, \boldsymbol{\xi}_t^z) < \underline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1}^z)$, add Benders optimality cut for Problem (19) at stage $t-1$.
    end for
  end if
end while
return $\underline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1})$ and $\overline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1}), \forall t \in \mathcal{T}$.

APPENDIX B
PROOF OF LEMMA 1

Proof of Lemma 1: We begin by establishing that $\underline{\mathcal{F}}_t$ is a polyhedral set. Initially, $\underline{\mathcal{F}}_T$ is a hypercube and thus polyhedral. Since the support set Ξ_T is a box uncertainty set, it has a finite number of vertices. For each vertex $\hat{\boldsymbol{\xi}}_T \in \Xi_T$, we have

$$\underline{\mathcal{F}}_{T-1}(\hat{\boldsymbol{\xi}}_T) = \big\{\boldsymbol{x}_{t-1} \in \mathbb{R}^n \;\big|\; \exists(\boldsymbol{x}_T, \boldsymbol{y}_T) \text{ such that } \boldsymbol{A}_T\boldsymbol{x}_{T-1} \geqslant \boldsymbol{h}_T + \boldsymbol{H}_T\hat{\boldsymbol{\xi}}_T - \boldsymbol{B}_T\boldsymbol{x}_T - \boldsymbol{G}_T\boldsymbol{y}_T, \; \boldsymbol{x}_T \in \underline{\mathcal{F}}_T\big\},$$

which is defined by finitely many linear inequalities and is therefore polyhedral. According to Claim 1 in [34], $\underline{\mathcal{F}}_{T-1}$ can be represented as the intersection of $\underline{\mathcal{F}}_{T-1}(\hat{\boldsymbol{\xi}}_T)$ over all vertices, which possesses a polyhedral structure. By repeating this argument recursively, $\underline{\mathcal{F}}_t$ can similarly be expressed as the intersection over finitely many vertices $\hat{\boldsymbol{\xi}}_t$, confirming that $\underline{\mathcal{F}}_t$ is polyhedral.

It is known that $\overline{\mathcal{F}}_t$ is the convex hull of a finite set and is therefore polyhedron. Since $\underline{\mathcal{F}}_t$ is polyhedral with bounded state variables, any points in $\underline{\mathcal{F}}_t$ can be expressed as a convex combination of the simplex listed in (18) of $\overline{\mathcal{F}}_t$. Hence, $\underline{\mathcal{F}}_t \subseteq \overline{\mathcal{F}}_t$. ■

APPENDIX C
PROOF OF THEOREM 1

The proof of Theorem 1, along with the supporting Lemmas 2 and 3, is provided.

Lemma 2: For each stage $t \in \{2, \ldots, T\}$, there are finitely many feasibility cuts generated from $\overline{\text{SP}}_t(\boldsymbol{x}_{t-1}, \boldsymbol{\xi}_t)$, which serve to refine the feasible regions $\underline{\mathcal{F}}_{t-1}$ and $\overline{\mathcal{F}}_{t-1}$ of the state variable $\boldsymbol{x}_{t-1}$.

Proof: Since the constraints of the bilinear programming Problem (21) define a polyhedral set, the optimal solutions $\boldsymbol{\pi}_t$ and $\boldsymbol{\xi}_t$ correspond to extreme points of the polyhedral and box uncertainty sets, respectively. As the number of extreme points is finite, the number of optimal solutions is also finite, and hence only finitely many feasibility cuts can be generated by $\overline{\text{SP}}_t(\boldsymbol{x}_{t-1}, \boldsymbol{\xi}_t)$. Therefore, for each stage t, the feasibility cuts are finite. ■

Different from RDDP under the RCR condition, infeasible points may cause the upper-bound problem to yield an invalid approximation and render the lower-bound problem infeasible, which hinders algorithmic convergence. Therefore, it is necessary to discuss the boundedness of the constructed lower and upper bound functions.

Lemma 3: (*Boundedness*) The proposed algorithm can ensure that the constructed lower and upper bound functions $\underline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1})$ and $\overline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1})$, for $t = 2, \ldots, T$, remain bounded.

Proof: We first establish that, for $\boldsymbol{x}_{t-1} \in \underline{\mathcal{F}}_{t-1}$, the function $\underline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1})$ is bounded. Assume the contrary, i.e., there exists $\boldsymbol{x}_{t-1}$ such that $\overline{\text{SP}}_t(\boldsymbol{x}_{t-1}, \boldsymbol{\xi}_t) > 0$. In this case, an additional feasible cut could be introduced to further refine $\underline{\mathcal{F}}_{t-1}$, which contradicts Lemma 2, as the number of feasibility cuts is finite.

Besides, we show that, for $\boldsymbol{x}_{t-1} \in \overline{\mathcal{F}}_{t-1}$, the optimal value of $\overline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1})$ is valid. From Lemma 1, we know that $\underline{\mathcal{F}}_{t-1} \subseteq \overline{\mathcal{F}}_{t-1}$. Since any point in $\overline{\mathcal{F}}_{t-1}$ can be expressed as a convex combination of the initial simplex, a feasible point $\boldsymbol{x}_{t-1} \in \underline{\mathcal{F}}_{t-1}$ naturally provides a valid bound for $\overline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1})$. Moreover, any infeasible point $\boldsymbol{x}_{t-1} \in \overline{\mathcal{F}}_{t-1} \backslash \underline{\mathcal{F}}_{t-1}$ is excluded by the introduced feasibility cuts, ensuring that the optimal value of $\overline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1})$ remains valid. This completes the proof of Lemma 3. ■

Proof of Theorem 1: We first establish that the forward pass always terminates after finitely many steps, and then demonstrate that both the upper and lower bound functions can be refined within finitely many iterations until convergence.

To prove the first claim, note that the forward pass returns to a previous stage whenever $\overline{\text{SP}}_t(\boldsymbol{x}_{t-1}, \boldsymbol{\xi}_t) > 0$. In such cases, a feasibility cut is incorporated into both the upper and lower bound problems. Since only finitely many feasibility cuts exist at each stage (Lemma 2), the condition $\overline{\text{SP}}_t(\boldsymbol{x}_{t-1}, \boldsymbol{\xi}_t) > 0$

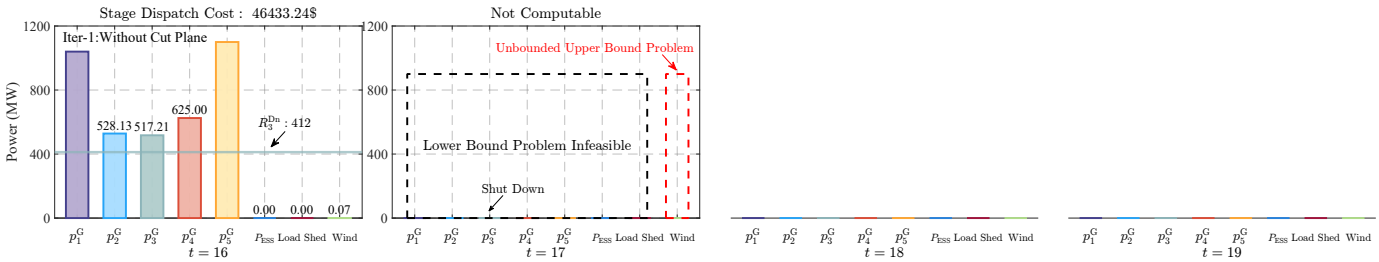
eventually becomes false. Consequently, the forward pass proceeds stage by stage and terminates at the final stage T.

We now establish the second statement. Since the lower-bound function $\underline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1})$ is bounded at each iteration (Lemma 2), and only finitely many optimality cuts can be generated, there exist only finitely many distinct lower-bound functions at each stage. Moreover, since the forward pass terminates after finitely many steps and only finitely many distinct lower-bound functions exist, the FRDDP algorithm can visit only finitely many candidate solutions [22] in the forward pass. Consequently, in the backward pass, each candidate solution yields a finite number of upper-bound points for refining the upper-bound function, implying that only finitely many distinct upper-bound functions $\overline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1})$ can be constructed.

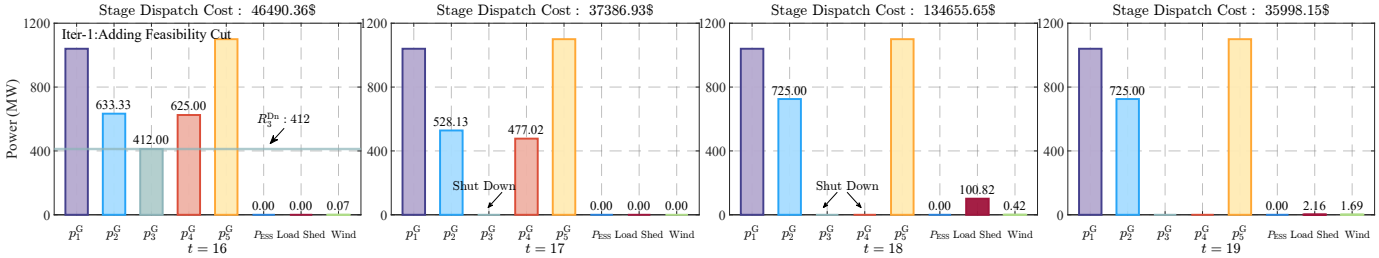
Since both $\underline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1})$ and $\overline{\mathcal{Q}}_t(\boldsymbol{x}_{t-1})$ are bounded and can be refined only finitely many times, they eventually remain unchanged as the iterations progress. Therefore, algorithm yields an optimal solution $\boldsymbol{x}_t$ such that $\underline{\mathcal{Q}}_{t+1}(\boldsymbol{x}_t) = \mathcal{Q}_{t+1}(\boldsymbol{x}_t) = \overline{\mathcal{Q}}_{t+1}(\boldsymbol{x}_t), \quad t \in \{1, \ldots, T\}$. ■

APPENDIX D
STAGE INFEASIBLE IN THE FIRST ITERATION

The occurrences of stage infeasibility during the first two iterations are illustrated in Figs. 7 and 8, respectively.

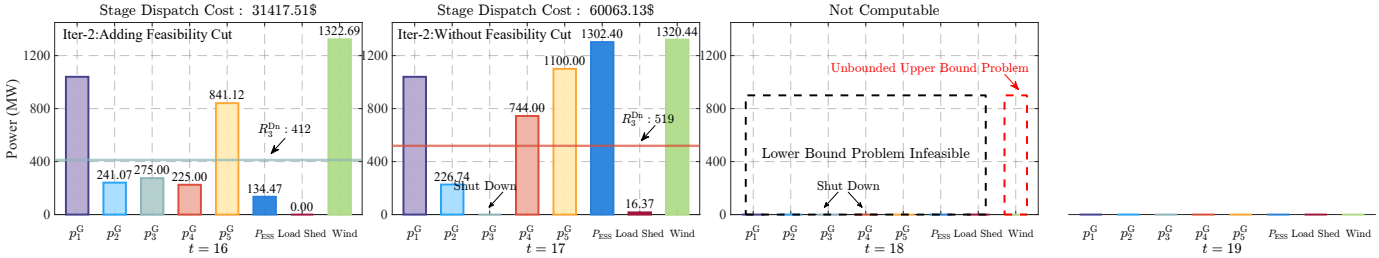


(a) The infeasibility led to the termination of the forward recourse (Generator 3 cannot be safely shut down at stage 17).

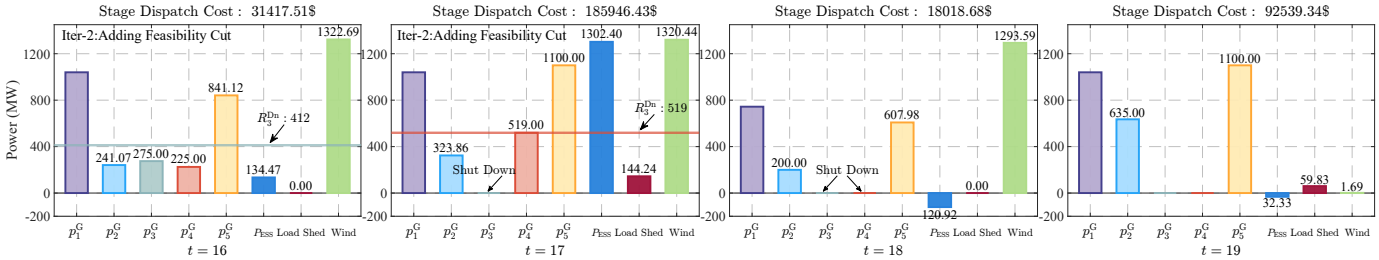


(b) Forward pass results after adding feasibility cuts at stage 16 (Generator 3 can be safely shut down).

Fig. 7. Scheduling results from stages 16 to 19 in the first iteration of training the FRDDP algorithm.



(a) The infeasibility led to the termination of the forward recourse (Generator 4 cannot be safely shut down at stage 18).



(b) Forward pass results after adding feasibility cuts at stage 16 and 17 (Generators 3 and 4 can be safely shut down).

Fig. 8. Scheduling results from stages 16 to 19 in the second iteration of training the FRDDP algorithm.